

Creating value through external intellectual property commercialization: a desorptive capacity view

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Abstract In open innovation systems, capturing value through external intellectual property (IP) commercialization is an increasingly important strategy for firms to keep pace with competitive changes. However, many firms have major difficulties in creating value through external patent exploitation. To understand these challenges, this paper explores how firms manage their external patent exploitation based on a multiple case study research design with fourteen firms from the pharmaceutical and chemical industry. Adopting a desorptive capacity perspective we find four main factors influencing the firms' management of external patent exploitation: the type of value creation, the organizational structure, the locus of initiative, and the extent of know-how transfer along with the patent. Based on these factors, three archetypes of external patent exploitation with different levels of desorptive capacity are identified. The article extends the concept of desorptive capacity and existing literature on intellectual property management in the context of open innovation. Managerial implications helping firms to implement external IP commercialization structures are discussed.

Keywords Intellectual property (IP) · IP commercialization · External patent exploitation · Licensing · Desorptive capacity · Case studies

JEL Classification L24 · O32 · O33 · O34

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1 Introduction

In the traditional closed innovation system, intellectual property (IP) rights such as patents were seen as a means to protect own technologies from being imitated (Chesbrough 2003). However, due to significant changes in the competitive environment, e.g., globalized markets, high dynamics in technology development (Bianchi et al. 2011a) and a growing mobility of highly skilled people (Florida 2002), closed innovation approaches become less viable and limit interactions with the firms' environment (Teece 1998). Many firms have recognized the potential of opening up their innovation processes and increasingly look for and share ideas with organizations outside of their firm boundaries (Gassmann 2006). From a patent management perspective, opening up the innovation process requires a shift from the traditional patent protection approach towards a wider approach that considers patents also as tradable assets. Within this context, firms have started to commercialize their patents externally and out-license and sell patented technologies (Davis 2004; Rivette and Kline 2000). This trend is underlined by the development of revenues from technology licensing over the last decades: From 1980 to 2003, the technology licensing payments and receipts have increased on average by 10.7 % per year (Granstrand 2004). Moreover, the value of technology transactions has grown from about \$40 billion in the late 1990s to an estimated \$100 billion in the early 2000s (Arora and Gambardella 2010). This development is argued to be due to the increasing competitive pressure through the globalization of markets, customers, and competitors which force firms to find ways to keep pace with the environmental changes and to increase their overall IP management effectiveness (Arora et al. 2001; Granstrand 2004). At the same time, recent studies claim a high level of unused patents within the firms. In a large scale study with European firms, Giuri et al. (2007) find that nearly 18 % of the patents are unused, in the pharmaceutical and chemical industry even 22 %. A US based survey revealed that more than 35 % of patented technologies are not used by the firms (Rivette and Kline 2000). A closer look at this licensing paradox of increasing technology licensing revenues on the one hand, and a large rate of underexploited patents on the other hand, implies that relatively few firms account for the majority of the increase in licensing, while many firms still engage little in out-licensing (Lichtenthaler and Lichtenthaler 2010). A recent study revealed that 40 % of the patent holders of all US patents account for 99 % of the US licensing revenues, while 60 % of the patent holder only generate 1 % of the licensing revenues (Alexy et al. 2009). In contrast to often cited examples such as IBM and Dow which have established comprehensive licensing programs and generate large amounts of money through externally commercializing their patents (Davis and Harrison 2001), many firms fail to exploit the full potential of their intellectual assets (Elton et al. 2002). This discrepancy points to a large heterogeneity between the firms and literature increasingly refers to managerial difficulties to implement external patent exploitation (Rivette and Kline 2000; Elton et al. 2002; Lichtenthaler 2011).

Several studies exist on the strategy of external patent exploitation, e.g., revealing that the firm's willingness to externally exploit its patents depends on the potential competitive advantage, the strategic appropriability, and the risk involved (Davis and Harrison 2001; Pitkethly 2001; Kline 2003). Studies also exist on the firms' motivation for external patent exploitation, including financial and strategic motives such as entering new markets and setting industry standards (Grindley and Teece 1997; Ehrhardt 2004; Palomeras 2007), and firms' organizational structures for technology trade (Granstrand 2004; Bianchi et al. 2011b; Lichtenthaler 2011). Also, the markets for patents have been investigated, e.g.,

revealing that asymmetric information and high transaction costs present barriers for patent commercialization (Gambardella et al. 2007; Benassi and Di Minin 2009; Teece 1998).

However, despite the practical importance of external patent exploitation and the fact that many firms experience substantial difficulties in managing it (Rivette and Kline 2000), literature lacks empirical insights in how firms organize and implement their external patent exploitation. Extant literature is mostly limited to conceptual approaches (Grindley and Teece 1997) and anecdotal evidence (Kline 2003; Rivette and Kline 2000). Moreover, most studies adopt knowledge or technology as the unit of analysis within exploitation activities, including the exploitation of unpatented technology. Based on the findings of several studies claiming that the firms' patent portfolios are often underexploited and hold significant unused commercial potential (Giuri et al. 2007; Rivette and Kline 2000; Elton et al. 2002), understanding external technology exploitation from a patent management perspective is an important gap in research. In particular, there is a lack of in-depth empirical insights into how firms actually manage their external patent exploitation. Consequently, this article aims to fill this gap by empirically investigating how firms manage external patent exploitation. Applying the concept of desorptive capacity as theoretical research setting (Lichtenthaler and Lichtenthaler 2009), we explore which factors influence the firms' approach towards managing external patent exploitation. The concept of desorptive capacity was introduced to complement the well-known concept of absorptive capacity and describes the firms' ability to externally exploit knowledge (Lichtenthaler and Lichtenthaler 2009). In the context of this article, desorptive capacity is understood as the firms' ability to externally commercialize their patents, in other words their "external patent commercialization capacity". In order to answer the research question, the empirical analysis of the paper is based on data from fourteen firms of the pharmaceutical and chemical industry.

The remainder of this paper is structured as follows. In Sect. 2, a brief overview of relevant literature on external patent exploitation and its role in the pharmaceutical and chemical industry is given. In Sects. 3 and 4, the research framework is developed and the research methodology is described. Sections 5 and 6 present and discuss the empirical findings and the developed archetype model. Lastly, Sect. 7 concludes the paper and outlines future research avenues.

2 External exploitation of patents and its role in the pharmaceutical and chemical industry

2.1 External patent exploitation

Intellectual property management is commonly understood as the management of a firm's IP rights consisting of patents, trademarks, industrial designs, and copyrights. In this study we focus exclusively on patents for two reasons: first, patents are the most tangible form of IP rights and enjoy the strongest legal protection compared to other types of IP. Second, patents mostly have, compared to the other IP rights, the greatest effect on a firm's commercial performance (Lerner 1994; Ernst 2001; Rivette and Kline 2000).

The intention of the patent system is to create incentives for investments in R&D and the diffusion of R&D results by enabling firms to appropriate returns from innovation (Teece 1986; Levin et al. 1987). To do this, literature distinguishes between internal and external exploitation of patents. Internal patent exploitation involves protecting own products and processes from imitation, creating entry barriers for competitors into a

technological field, and ensuring freedom to operate, i.e., creating and maintaining a position where the firm can continue its research and development (R&D) activities free from third party IP rights (Levin et al. 1987; Harabi 1995; Cohen et al. 2000; Blind et al. 2009). External patent exploitation refers to the exploitation of patents outside the firm boundaries through out-licensing, cross-licensing, or selling of patents (Granstrand 2000; Monk 2009). To clarify the meaning of the term ‘external patent exploitation’, we define it, based on the definition of external knowledge exploitation of Lichtenthaler (2005), as follows: *External patent exploitation describes an organization’s deliberate exploitation of patents to another independent organization with or without know-how transfer involving a contractual obligation for monetary or non-monetary compensation.*

2.2 The case of the pharmaceutical and chemical industry

The important role of patents in the chemical and pharmaceutical industry has been subject of various studies (e.g. Teece 1998; Arora et al. 2001; Gambardella 1995). The character of pharmaceutical and chemical inventions enables firms to effectively define patent rights over a specific chemical composition and to create temporary monopolies (Levin et al. 1987). Cohen et al. (2000) have termed this characteristic as ‘discrete technologies’, i.e., technologies where one patented substance often protects one product. This is in contrast to, for example, the electronic industry where final products are protected by entire patent clusters. As a result, research and development of pharmaceutical and chemical firms is more dependent on patents than most other industries (Thumm 2001; Granstrand 2000).

Furthermore, pharmaceutical and chemical firms are under increasing regulatory and commercial pressure due to exceptionally high initial R&D costs (Whitehead et al. 2008; Arora and Fosfuri 2000) and long product development life cycles (Teece 1998). In the pharmaceutical and chemical industry, the average time until a new product is launched is between 7 and 15 years (Fine 1996). Considering the maximum patent life time of 20 years, with some exceptions in the pharmaceutical industry with 25 years, firms often have only a short time to realize earnings from their products. Moreover, the fact that only about one out of 10,000 substances discovered in the R&D phase in the pharmaceutical industry becomes a marketable product (Gassmann and Reepmeyer 2005) shows the tremendous risk that these firms face regarding the outcome of their innovation efforts. Towards this backdrop, pharmaceutical and chemical firms need to look for improved commercialization opportunities and increasingly adopt a more proactive patent management approach by capturing additional value through externally exploiting their patents (Arora and Fosfuri 2000).

Despite the increasing importance of external patent exploitation especially for the pharmaceutical and chemical firms, literature lacks empirical insights into how the firms actually manage external patent exploitation. This article aims to fill this gap by empirically investigating which factors influence the firms’ management of their external patent exploitation and how this impacts the success thereof. In order to analyze the data, a research framework adopting the concept of desorptive capacity is developed in the subsequent section.

3 Research framework

Open innovation literature distinguishes between outside-in and inside-out flows of knowledge at firms (Gassmann et al. 2010). The outside-in process entails acquiring and

integrating external knowledge into the firm. The inside-out process implies the transfer of own knowledge assets outside to third parties, e.g., through licensing or selling of technologies. The dynamic capability theory assumes that organizations need to adopt to changing business environments and renew their competences in order to stay competitive (Teece et al. 1997). Thus, firms in open innovation systems need to develop capabilities to internalize and externalize knowledge to gain competitive advantage. Regarding the outside-in process, the widely known concept of absorptive capacity has been successfully applied in numerous studies. The theory of absorptive capacity was originally introduced by Cohen and Levinthal (1990) and refers to a firm's ability to recognize, evaluate, assimilate, and apply external knowledge. While the outside-in process and the firm's absorptive capacity have been comprehensively studied in previous research, its counterpart, the inside-out process, is still under-researched. Towards this backdrop, Lichtenthaler and Lichtenthaler (2009) recently introduced the concept of *desorptive capacity*. As opposed to *absorbing* external knowledge, *desorptive* capacity refers to a firm's ability to externalize internal knowledge assets in order to appropriate returns from innovation. Accordingly, Lichtenthaler and Lichtenthaler (2009, p. 1321) define desorptive capacity as "a firm's ability to externally exploit knowledge".

The authors propose two main stages of desorptive capacity within the external technology exploitation process: the *identification* of technology transfer opportunities and the *transfer* of the technological knowledge. According to the authors, identifying potential technological assets for being externally exploited is an often underestimated but crucial managerial challenge for the firms. Many firms face difficulties in recognizing external exploitation options due to the lack of transparency of technology markets (Lichtenthaler and Lichtenthaler 2009). At the same time, a study shows that a proficiently managed process to identify new external exploitation options positively impacts a firm's external technology commercialization performance (Lichtenthaler et al. 2009). Therefore, developing capacities to initiate an external exploitation project is a key aspect of developing desorptive capacity. In the second stage, the transfer stage, the firms have to assign a value to the patents and to develop a pricing strategy considering the cost benefit ratio. Also, the contractual agreement with the partner firm is negotiated. Besides the financial compensation this can also include the agreement on additional know-how to be transferred to facilitate the application of the technology at the partner firm.

Furthermore, dynamic capability theory suggests that the transfer of knowledge is affected by *internal* and *external* variables (Lichtenthaler et al. 2010), which we, based on prior research on knowledge and technology transfer, consider for our research framework regarding external patent exploitation.

3.1 Internal variables

3.1.1 Motives

Studies on the firms' rationales to externally exploit their IP suggest two major types of motives: monetary and strategic motives. Monetary motives have the objective to enable direct revenues and include generating new income, maximizing the return of investment, and reducing maintenance costs (Palomeras 2007). Strategic reasons entail entering new markets, maintain freedom to operate, setting industry standards, create a complementary relationship with partners, gain access to third party IP, reducing risks, and achieve learning effects (Kutvonen 2011; Koruna 2004; Lichtenthaler 2005) and allow the firms to generate long-term indirect revenues. Previous findings in literature suggest that the motive

of external technology exploitation has an influence on how firms manage the exploitation process. For example, Lichtenthaler (2007) states that strategic driven technology licensing is much more complex than purely monetary driven licensing and hence requires a different managerial approach.

3.1.2 Organization

A firm's organizational structure also impacts its management of external technology exploitation. Extant literature emphasizes two perspectives on the firms' organization of IP management. On the one hand, the way how IP management is implemented within the firms' organizational structure differs between a centralized versus a decentralized IP structure (Carlsson et al. 2008; Granstrand 2000). On the other hand, literature points to the importance of dedicated versus temporary IP functions. In a study with 152 firms, Lichtenthaler et al. (2009) highlight the role of dedicated and temporary organizational structures in the early technology exploitation phase, i.e., for identifying new external exploitation opportunities. Also, in a multiple case study analysis with firms from the pharmaceutical industry, Bianchi et al. (2011b) find that the firms' organization of external technology commercialization is linked to distinct characteristics of the exploitation project. They find that adapting the organizational structures according to the strategic relevance of the project, the volume of the transaction, and the maturity of the technology can be an important success factor of external technology commercialization.

3.2 External variables

3.2.1 Technology market

Towards the backdrop of the open innovation era, the role of patents has shifted from solely protecting a product or a process from imitation towards being a tradable good itself. In this vein, markets for patents and technologies have emerged (Arora et al. 2001; Guilhon 2001; Teece 1981). However, markets for intangible goods such as patents differ significantly from product markets. Although studies have observed a growth of technology markets and licensing revenues over the last decades, they also point to a major aspect that limits the growth of these markets: high transaction costs caused by asymmetric information between the two involved parties and uncertainty, e.g., regarding the extent and legal status as well as the economic value of the patent (Arora and Gambardella 2010). Many firms do not finish or enter a licensing deal at all because of high transaction costs such as costs for finding a partner, negotiation effort, or a weak legal strength of the patents (Arora and Gambardella 2010; Gambardella et al. 2007). In a survey with US and Canadian firms Razgaitis (2004) found that only for 25 % of licensable technologies a partner firm could be found, and that again out of these, only 25 % entered into negotiations of which then about half of the deals were closed. Along with the rise of the market for patents and technology and the firms' difficulties to act on these markets, intermediary services have emerged (Ruether 2012; Ewing and Feldman 2012). IP intermediaries can be described as "organizations whose business is to match supply and demand of technology to facilitate IP-based transaction" (Benassi and Di Minin 2009, p. 69). Their activities entail for example advising the firms in evaluating their patents, finding new business options, developing business models, and identifying potential partner firms.

3.2.2 Partners

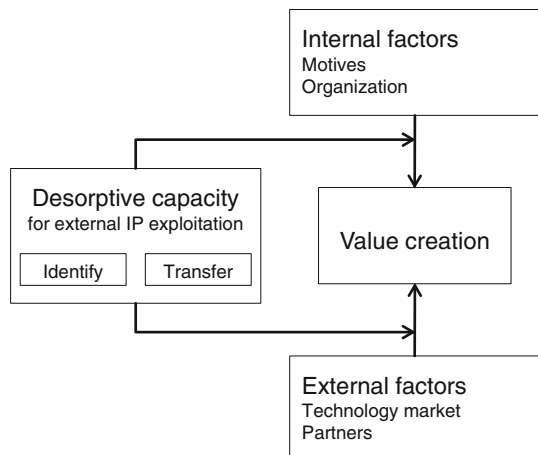
Besides the patent owning firm's exploitation strategy and motives, the extent of the patent and technology transfer also depends on the partner firm's previous know-how and competencies of the technology, or, in other words, on the partner firm's need for additional know-how to apply the technology. In this regard, literature distinguishes between external patent exploitation with and without the transfer of know-how (Arora and Gambardella 2010). External patent exploitation without know-how transfer stands for the transfer of the bare legal right and is often used to solve the freedom to operate issues (Pitkethly 2001). External patent exploitation with know-how transfer includes the transfer of related explicit and implicit knowledge such as research reports and technological expertise. These transactions require an active participation of the patent owner to facilitate the successful application of the technology at the partner firm (Lichtenthaler and Lichtenthaler 2010).

In Fig. 1, the above presented aspects are synthesized in a research framework on which the subsequent data analysis will be based. Building on the concept of desorptive capacity (Lichtenthaler and Lichtenthaler 2009, 2010), we consider both internal and external factors impacting the firms' management of external patent exploitation, which, in turn, is assumed to impact leveraging and capturing value of the firms' patent portfolios.

4 Research methodology

Due to the scarce empirical insights pertaining to how firms manage their external patent exploitation, the research is based on a qualitative, exploratory research design. A multiple case study research with the firm's external patent exploitation activities as the unit of analysis is employed. This research design allows addressing detailed questions to gain deeper insights in the firms' patent management approach and to understand the context and motivation of behavioral patterns (Yin 2009; Eisenhardt 1989; Gibbert et al. 2008). The cases were selected in a two-step selection process following theoretical sampling (Eisenhardt 1989). In a first step, initial interviews with 50 firms following Yin's replication logic (Yin 2009) were carried out in order to gain general insights into the firm's external patent exploitation activities. In a second step, fourteen firms were chosen which

Fig. 1 Research framework



provide the most detailed and insightful information and have the highest learning potential with respect to their external patent exploitation approach (Eisenhardt 1989). First, we followed the contingency approach which suggests that an organization's ability to achieve its goals depends on the organizational situation and context (Fiedler 1964). In the context of external patent exploitation, we focus on the industry as a contingency factor (Mansfield 1986; Chenhall 2003) and exclusively investigate firms from the pharmaceutical and chemical industry. With regards to patent management, these industry sectors are highly comparable because they deal with discrete technologies (Cohen et al. 2000). Second, we included firms representing 'polar types', i.e., illustrating extreme sides of the investigated phenomenon (Yin 2009). To do this, we chose firms with both high and low activities of external patent exploitation. Third, we considered the firms' accessibility and chose those firms which allowed for a personal face-to-face contact. Also, due to the sensitive data on the firms' intellectual property strategies, building up a trust relationship through personal contact is a crucial aspect for the quality of the data. As a result, seven firms headquartered in Germany, six in Switzerland, and one in the United States of America were chosen.

The data collection was conducted through personal face-to-face and telephone interviews using a semi-structured interview guideline. For the initial interviews, the heads of the firms' IP departments were contacted. For the interviews with the fourteen selected firms, at least one more detailed follow-up interview was conducted with the same persons contacted for the initial interviews. Whenever possible, one or two other employees of the firms were interviewed. This was the case for nine of the fourteen firms: With seven firms, a second person could be interviewed, with two firms two other persons. While the first interview partner at all firms was the head of IP, the other interviewed employees where IP and licensing managers involved in external patent exploitation projects (see Table 1). In addition to the interviews, data was collected through desk research analyzing the firm's internal and external documents and press releases. This multiple source approach allows to triangulate the data which increases the validity and reliability of the results (Jick 1979).

The fourteen sample firms are all research-intensive pharmaceutical and chemical firms. The size of the investigated firms ranges from 80 to more than 100,000 employees, and their patent portfolios contain between 450 and 140,000 patents. To start the data analysis, we synthesized all data into individual case histories. Each case history includes narratives, selected quotes from the interviewees, and tables summarizing the key facts. Then, within-case analysis was used to identify the specific way how each firm manages their external patent exploitation (Eisenhardt 1989). When all individual case histories were completed, the cross-case analysis was conducted to identify general patterns across the cases. To do this, the data was tabulated and cross-case pair wise comparisons were conducted (Miles and Hubermann 1994). This allowed us to see similar themes across the cases and to deduce factors that impact the firms' management and approach towards external patent exploitation.

5 Findings and discussion

5.1 Internal factors influencing the firms' external patent exploitation

5.1.1 *Motives of external patent exploitation*

The motive why firms exploit their patents externally often determines the form of external exploitation, i.e., out-licensing, cross-licensing, and the sale of patents. The firms in our

Table 1 Overview of investigated firms

Firm	Headquarters	Employees	Interviewee(s)	Interviews
Firm1	Germany	>80,000	Head of IP (regional), IP Manager	2
Firm2	Germany	<500	Head of IP, IP Portfolio Manager	2
Firm3	Germany	>1,000	Head of IP	2
Firm4	Switzerland	<500	Head of IP, Licensing Manager	4
Firm5	Germany	>30,000	Head of IP, IP and Licensing Manager	2
Firm6	Germany	>40,000	Head of IP, IP Manager	2
Firm7	USA	>10,000	Head of IP (regional)	2
Firm8	Switzerland	>10,000	Head of IP, IP Manager	2
Firm9	Germany	<500	Head of IP	2
Firm10	Switzerland	>80,000	Head of IP (pharma), IP Manager (corporate), IP and Licensing Manager (pharma)	5
Firm11	Switzerland	>80,000	Head of IP (corporate), Head of IP (regional), Licensing Manager	5
Firm12	Switzerland	>10,000	Head of IP, IP Manager	4
Firm13	Germany	>5,000	Head of IP	2
Firm14	Switzerland	>20,000	Head of IP	2

study use all of these forms and follow both financial and strategic motives to commercialize their patents outside of the firm boundaries. Generating revenues and cutting costs are the most important *financial* motives at the investigated firms. Firm1, for example, sells patents of terminated research projects to reduce the maintenance costs of its patent portfolio and to receive additional revenues. Selling a patent leads to a one-time revenue and shifts the patent ownership to the partner firms. In contrast, an out-licensing agreement usually consists of an initial up-front payment and subsequent regular royalty payments. Firm11, for example, has out-licensed a terminated research project to a spin-off company and has gained constant revenue streams through the royalty payments of 10 % of sales for more than a decade.

Strategic motives to externally exploit patents are more diversified than financial motives. One of the most important strategic motives is the transfer of technologies for further development and product commercialization. Firm3, for example, out-licenses or sells patents of secondary technologies which need further development to finalize a product that exceeds Firm3's own resources. Another important strategic motive is to gain access to third party patents. Firm6 uses cross-licensing agreements to access patents of other firms or institutions which are important for their own already realized products or for potential future products. Moreover, for certain technologies, standardization can be a reason for external patent exploitation. Firm12, for example, considers to broadly out-license a sub patent portfolio containing approximately 40 patent families covering a specific technology in order to set an industry standard. The technology has the potential to

being widely used within the own industry as well as in applications in other industries. Moreover, through out-licensing either in broad licensing programs or exclusively to selected partner firms, the firms also intend to enhance the firm's reputation. Their objective is to be perceived as an innovative and technology creating firm. Finally, out-licensing for legal reasons is a motive at the analyzed firms. While none of the firms follows an aggressive approach regarding patent enforcement, they are willing to defend their patents in case of infringement. Firm7, for example, uses out-licensing and cross-licensing agreements to solve infringement cases. The additional revenue is used to further innovate and develop new products.

From a desorptive capacity perspective, the motive for external patent exploitation refers to the type of value the firm wants to create. The firms approach towards external patent exploitation is closely related to their business strategies and commercialization capabilities (Teece 1986). While some of the sample firms treat out-licensing as a distinct opportunity to gain additional revenues and follow a rather short-term perspective, others intend to engage in long-term alliances with its partners and build up a network for future out-licensing opportunities. Respectively and in line with Chesbrough's argumentation on creating value through managing IP, the firms follow different business models and have defined different strategies to appropriate returns through their patents portfolio (Chesbrough 2003). The role of the firms' business model can be demonstrated with the example of Firm2 which changed its business strategy over the last years including an active shift of its approach towards patent exploitation. In the past, Firm2 regarded its patents as a 'safety fence' to protect products and block competitors. Only the internally used patents were considered as valuable patents and there was not any willingness to share patents with third parties. Today, Firm2 treats its patents as tradable assets with the objective to gain access to new technologies, create partnerships and joint ventures, and gain revenues. Accordingly, Firm2 restructured its patent management to better handle the external patent exploitation. Hence, firms are advised to establish desorptive capacities according to their IP commercialization strategy that defines if the firm aims to exploit its patents internally through protecting own products, externally through e.g., out-licensing, or a combination of both. Firms should define which type of value they wish to create through the external exploitation of the patent and adopt their management accordingly. This also includes decisions on which form of external exploitation, i.e., out licensing or selling, should be chosen. For out licensing, firms should decide whether the technology should be spread using non-exclusive licenses, or if the best benefit is reached by licensing it exclusively to one partner firm. While broad licensing can be appropriate to introduce an industry standard, exclusive licensing can be applied for early technologies that need further development.

5.1.2 Organizational structure

Based on our framework we analyzed the firms regarding their organizational structure of external patent exploitation considering centralized versus decentralized structures and dedicated versus temporary functions. In our sample, nine of the firms have a centralized and five a decentralized IP organization. This organizational difference seems to be independent of firm size. At Firm14, IP management is a decentralized function organized in multifunctional teams consisting of patent attorneys, legal attorneys, and licensing specialists. These teams act as interfaces between the R&D department and the business units and provide support during the creation of IP and the subsequent transformation thereof into intellectual capital. They also support decisions regarding the continuation and

strategic orientation of projects by providing the relevant information regarding the legal situation and state of the art of the technology to ensure freedom to operate. At Firm3 and Firm12, the IP function is organized centrally in a holding firm. The IP holding is the owner of all IP rights of the two firms. At Firm12, the IP holding is financed by licensing patents internally to own companies and externally to third parties. While currently most of the licensing deals are still conducted internally, Firm12 aims to enhance the exploitation of patents outside of the firm.

However, with regards to the firms' desorptive capacity, we did not find a significant pattern between firms having centralized or decentralized patent functions. Instead, supporting the findings from Bianchi et al. (2011b) and Lichtenthaler (2011), we find that dedicated or temporary functions for external patent exploitation impacts the firms' management of external patent exploitation and their desorptive capacity. In our sample, eight of the firms have dedicated functions for external patent exploitation. Firm5, for example, has established a technology transfer and licensing department exclusively responsible for the externalization of its intellectual assets. However, our findings suggest that a dedicated external patent exploitation function is not always the best solution. Firm11, for example, has had a dedicated function but recently reduced it and integrated it into the patent acquisition function because of the low frequency of external patent exploitation projects. Hence, especially firms focusing on product commercialization with limited external patent exploitation activities should consider the cost benefit ratio of a dedicated function and assess which desorptive capabilities they need and can afford. For them, a well-organized project based temporary team staffed with experts related to the respective technology might be the more efficient way to manage the external exploitation project.

Despite these differences regarding the organizational structure, all firms have in common that the teams managing the external patent exploitation—within a dedicated or a temporary function—are highly multidisciplinary. Besides patent specialists there are employees from the R&D, business development, legal, and marketing department involved to decide whether a patent should be externalized or not and to prepare and realize the transfer of the patent. This is in line with Bianchi et al. (2011b) who find that for both dedicated and temporary functions employees with heterogeneous skills are involved. For example, within their sample firms the dedicated functions are composed of at least six experts from different disciplines.

The multidisciplinary nature of external patent exploitation is seen by the firms both as a challenge to coordinate the different perspectives of e.g., the IP, R&D, marketing, and legal departments, but most importantly as necessary in order to manage the complexity of the external patent exploitation. Thus, firms should move away from regarding patent management as a pure legal function and increase the awareness of the multidisciplinary nature of external patent exploitation to increase the chances of success. This involves, for example, to ensure that all needed information for an external exploitation project can be accessed by the external exploitation function, e.g., at the R&D, marketing, and business development department. This requires a general openness and commitment at the firm to use the patents beyond the firm's traditional boundaries (Chesbrough 2003; Kline 2003), or differently put, to overcome the not-sold-here attitude (Lichtenthaler et al. 2010).

5.2 External factors influencing the firms' external patent exploitation

5.2.1 Partners

The requirements of the partner firms with regards to additional know-how besides the patent itself as well as their expectations regarding the quality of the patent significantly

impact the management of the external patent exploitation at the investigated firms. To ensure the quality and to meet the expectations of its partners, Firm2, for example, bases the internal selection of tradable patents on specific quality criteria. The technology must be internally tested through a feasibility study and prototypes, and the documentation of the technology must be available. Also, Firm2 has an internal policy that the remaining life time of the patent should be more than 6 years to allow the partner firm a reasonable period for commercialization. Due to its earlier experiences, Firm2 avoids out-licensing or selling patents of unsuccessful research projects or to trade solely the patent rights. The finding that the management of external patent exploitation depends on the extent of know-how transfer in addition to the pure patent right can be compared with Granstrand's (2004) conceptualization linking a firm's technology exploitation strategy with the degree of organizational integration of the technology at the receiving firm. A partner firm that requires comprehensive additional know-how probably intends to integrate the technology to a higher degree than a partner firm that only in-licenses the pure patent right. Hence, one can conclude that the management of external patent exploitation at the patent owning firms depends on the organizational integration of the patent at the receiving firm. Interestingly, the analysis of the firms has shown that all firms usually have at least a medium extent of know-how transfer when they commercialize their patents externally. According to the firms, offering a package of technology together with the patent is a key success factor of external patent exploitation. Firm2, for example, offers its partners comprehensive reports of the technology development. Also Firm3 mainly conducts out-licensing deals with know-how transfer providing documents and personal support through an R&D employee. Firm4 often cooperates with its partners for several years because the firm generates patents of early technologies which are not yet ready to be commercialized and need further development. Therefore, the transfer of technical know-how in addition to the patent is a core aspect for Firm4 to commercialize its patents. While literature on external technology and knowledge exploitation implicitly includes the transfer of know-how, but also considers the exploitation of unpatented technologies (Lichtenthaler 2009; Arora et al. 2001), literature approaching the topic from a patent management perspective mostly neglects the aspect of know-how transfer along with the patent (Rivette and Kline 2000; Kline 2003). Only Chesbrough (2003) underlines that the successful commercialization of patents should be treated as a business case and requires a business model. Hence, our results contribute to patent management literature by emphasizing the importance of offering additional know-how along with the patent transfer. To increase the success of external patent exploitation, firms should consider to provide a technology package rather than the sole patent right. This makes it easier for the partner firm to integrate the patent into its technology portfolio and to realize and market the final product successfully. Selling or licensing the pure patent right is mainly appropriate to solve freedom to operate issues.

5.2.2 Technology market

The target market for an external patent exploitation can either be within the firms' own competitive environment or outside of it. Especially when the patent is out-licensed within the own industry, the firms conduct a comprehensive analysis of the competitive situation. This helps on the one hand to assess the potential of the technology in the market, and on the other hand, to reduce their risk of unintended losses of their competitive position (Kline 2003). Out-licensing into another industry involves less risk regarding the firm's own competitive position and can be a source for substantial additional revenue (Elton et al.

2002). Firm12, for example, intends to out-license a technology, but considers licensing it first to firms in other industries because it is less risky with regards to the own competitive position than within its direct competitive environment. Before out-licensing the technology within the own industry, Firm12 intends to conduct a more comprehensive risk analysis.

Furthermore, the choice in which market the patent is transferred also influences the firms' willingness to cooperate with IP intermediaries. From a desorptive capacity perspective, cooperating with an IP intermediary can be an alternative to build own desorptive capacities. IP intermediaries can complement the firms capacities by helping the firms to find a partner firm and to prepare and realize the transaction (Benassi and Di Minin 2009). Our interviewees state that their main reason for working with an IP intermediary would be to complement the limited resources and realize a fast transfer of patents without huge internal effort. Firm8 for example, considers working with intermediaries to realize a fast commercialization of abandoned patents. At Firm2, the identification of a transaction partner sometimes requires the support of IP intermediaries, e.g., for the commercialization of patents in specific geographical regions such as China. The Chinese market is nearly unknown to Firm2. Additionally, cultural differences play an important role for a successful patent transfer deal. Therefore, for transferring patented technologies into Chinese markets Firm2 mandates IP intermediaries. In a study on the importance of intermediary services in the market for technologies, Lichtenthaler and Ernst (2008) argue that IP intermediaries should be seen as a complement for internal competencies. Hence, cooperating with IP intermediaries can help firms to complement and extend their own desorptive capacities of external patent exploitation. However, some efforts at the sample firms to close a deal with the support of an intermediary have failed because internally required resources were still higher than expected. Thus, while intermediaries should be considered as a supporting option for certain external patent exploitation projects, the firms should carefully assess which tasks could be done by an intermediary and which still have to be accomplished internally. Also, the intermediary should be chosen according to his competencies which best complement the firms' own desorptive capacities.

5.3 Internal-external factor: locus of initiative

The analysis of the firms reveals that another factor which was not considered in our research framework impacts the firms' management of their external patent exploitation: the locus of initiative of the external exploitation. An external patent exploitation project can be initialized either internally by the firm itself or externally by a third party. In case of an *external initiation*, a third party approaches the patent owning firm and makes a request to access certain patents. For the patent owning firm, the external patent exploitation process starts with assessing the value of the patents on which the keep-or-sell decision and the potential price for the transfer is based. Firm8, which usually out-licenses based on external requests, decides from case to case if the deal could be interesting. Especially licensing requests from firms in other markets are promising options for Firm8. In case of an *internal initiation*, the management process starts much earlier with identifying potential patents for external exploitation. At the investigated firms, a starting point often is a regular, e.g., annually, patent portfolio review. Based on the result of the periodical review, Firm12, for example, creates an exploitation portfolio containing patents for potential external exploitation. The exploitation portfolio allows Firm12 to structure the potential patents for external commercialization and to offer packages of patents to the partner instead of only single patents. The patents in the exploitation portfolio are

evaluated with regards to their commercialization potential. The evaluation is based on market data and the technology life cycle. Another major challenge of the self-initiated approach is to identify a partner firm. At our sample firms, the technologies are often quite specific which makes it difficult for the firms to find a suitable partner who can potentially use the patent. Proceeding with the example of Firm12, the firm creates in a first step a list with possible transaction partners. The list is based on experience and previous contacts. In a second step, Firm12 conducts patent searches to identify potential partners. To do this, patent data bases are analyzed according to patent classifications. For this patent search Firm12 cooperates with an IP service provider because this task would exceed Firm12's internal resources. The results of the patent search help Firm12 to find firms that might be interested in the technology because of related technologies and applications.

For the firms the active, internally initiated approach implies a lot more effort in the pre-commercialization phase than in case of an external initiative where a third party approaches the firm and asks for certain patents (Kutvonen et al. 2010). While in the externally initiated case the patents and partners are already known, an internally initiated exploitation project requires efforts to identify a potential patent as well as to find a suitable partner firm. The importance of these activities has been highlighted in several studies, especially the identification of a partner firm is considered a major challenge or even a barrier of external patent exploitation (Arora et al. 2001; Palomeras 2007). Thus, moving from a passive towards an active approach is a big step for every firm. Firms who consider shifting their external patent exploitation towards an active approach should especially pay attention to establishing capacities to manage the potential patent and partner identification.

6 Three archetypes of external patent exploitation

Based on the above discussion, four main factors influencing the external patent exploitation management at the investigated firms emerge. Firstly, based on the motive why the firm wants to externally exploit its patent the *type of value* the firm intends to create through the external exploitation influences the way it is managed. Secondly, the *organizational structure* at the firm is important. The findings suggest that a centralized or decentralized IP structure does not have a direct effect on the management of the external patent exploitation. Instead, it seems that the fact if and to what extent the firms have established dedicated functions for external patent exploitation is crucial. Thirdly, the *locus of initiative* of the external patent exploitation project determines at which point of the process the firm actively starts to manage the external exploitation. Fourthly, the requirements of the partner firms regarding additional know-how along with the patent has shown to be an important aspect the firms have to consider when out-licensing or selling their patents. Thus, the *extent of know-how transfer* from the patent owning firm to the partner firm is a factor impacting the firms' external patent exploitation management. The combination of different characteristics of these factors lead to three distinct archetypes of firms adopting different modes towards their management of external patent exploitation (Table 2). In the following, the three archetypes are described.

6.1 Ad hoc mode

The business strategy of the firms adopting the ad hoc mode is focused on product commercialization. Thus, their IP strategy is optimized to exploit their patents internally in own

Table 2 Three archetypes of external patent exploitation

Level of desorptive capacity	Archetype	Investigated firms	Type of value creation	Organizational structure	Locus of initiative	Extent of know-how transfer
High	Leverage mode	Firm4, Firm2, Firm9	Strategic + financial long-term perspective	All functions focused on external exploitation	Internal active, professionalized	Very high: comprehensive know-how transfer, often lasting for months or years
Medium	Hybrid mode	Firm3, Firm5, Firm7, Firm12, Firm14	Strategic mid to long-term perspective	Dedicated function, separated from acquisition function	Internal active	High: intensive but limited know-how transfer
Low	Ad hoc mode	Firm1, Firm6, Firm8, Firm10, Firm11, Firm13	Financial short term perspective	No permanent dedicated function, often combined with acquisition function	External reactive	Medium: short know-how transfer phase

products and processes. External patent exploitation is not the focus of their IP management and therefore occurs only on an ad hoc basis and is mostly initiated externally by firms that wish to use certain patents. The patent out-licensing and sale projects are mainly conducted to generate additional income or recoup costs. Hence, these firms basically aim to create a financial value through external patent exploitation. In our sample, the only exception is Firm6, who does not intend to generate revenue through licensing in the first place but engages in cross-licensing to access third party patents emphasizing the mutual exchange without a financial compensation. Due to the small role external patent exploitation plays for the business of these firms, they do not have a permanent dedicated function. The teams conducting the external exploitation are either staffed temporary for each project or the external exploitation function is coupled with the patent acquisition function.

6.2 Hybrid mode

The hybrid mode archetype includes firms following a twofold IP strategy. The firms consider both internal and external exploitation as an equal means to generate benefits of their patent portfolios. The main motive of external patent exploitation is often of strategic nature such as entering new markets difficult to access through a product strategy, introducing an industry standard, or transferring further development and production to a partner firm because of limited internal resources. An example for the hybrid mode is Firm3. Firm3 is a developing and producing firm for specialty chemistry. During the development of new products, Firm3 cooperates closely with partner firms leading to two options to exploit its patents. The first option is to use the patents internally in own products. The second option is that the partner firm takes over the production and the patents are exclusively out-licensed. In this case, the firm emphasizes to offer ‘licensing packages’ which, besides the patent itself, include the transfer of know-how for the integration and application of the technology at the partner firm. In contrast to firms following an ad hoc mode, the hybrid firms actively initiate the external patent exploitation themselves. Firm5, for example, estimates that roughly 95 % of their external patent exploitation projects are self-initiated, while only 5 % are initiated externally. Regarding the organizational structure, the firms in the hybrid mode have established dedicated external technology exploitation functions which complement the classical product commercialization.

6.3 Leverage mode

The firms following a leverage mode use external patent exploitation as their business strategy. Hence, they aim to create both strategic and financial value to ensure the survival of the firm. Respectively, the organization and the process management are tailored to external patent exploitation and are highly professionalized. In our sample, Firm4 adopts the leverage mode. Firm4’s business strategy focuses on the research intensive part of the pharmaceutical value chain. Thus, the firm aims to develop drug candidates until clinical phase two and then tries to partner with established pharmaceutical firms for late-stage development and marketing of the product. This strategy requires a comprehensive technology transfer phase in which the transfer of know-how is as important as the patent itself. The firms following the leverage mode often cooperate with their partners for months or even years to further develop the technology and realize the final product. Thus, especially

for the firms adopting the leverage mode, external patent exploitation involves a long-term commitment regarding their own resources and the collaboration with their partner firms.

From a desorptive capacity perspective, these three archetypes are characterized by different levels of desorptive capacity. The highest level of desorptive capacity can be found at those firms adopting the leveraging mode. Based on their long-term strategy to generate strategic and financial benefits, they have organized their daily business accordingly and all functions are focused on the externalization of the patented technologies. In contrast, the firms in the hybrid archetype consider external patent exploitation as a complement to their product commercialization. While they, equivalent to the leverage archetype, actively initiate external exploitation, the external patent exploitation function is not the focus but a separated part of their business. Firms using external patent exploitation on an ad hoc basis have the lowest level of desorptive capacity. Focus of their business is clearly the product commercialization and revenues through external patent exploitation rather a side effect. This finding is also in line with discussions on the role of earlier experience for building dynamic capabilities. Equivalent to absorptive capacity, Lichtenthaler and Lichtenthaler (2010) suggest that a firm's desorptive capacity is path-dependent, i.e., a firm's previous engagement in external patent exploitation influences its probability and success thereof (Teece et al. 1997; Kim and Vonortas 2006). Firms using external patent exploitation more frequently are more experienced and thus able to build a higher level of desorptive capacity than firms which rarely conduct external exploitation projects. To profit from their experience, firms should carefully analyze earlier patent exploitation projects and use their lessons learned to increase the chance of success for future projects. For example, one lesson learned at Firm2 was that many deals were not closed because it turned out that the expectations of the supply and demand side did not meet. To avoid this for future projects, Firm2 now pays attention that the expectations of both sides are clearly defined and communicated right from the beginning.

Furthermore, it should be noted that reaching a high level of desorptive capacity is not necessarily the most effective way to manage external patent exploitation. On the contrary, as the example of Firm11 in Sect. 5.1.2 demonstrated, firms using external patent exploitation on an ad hoc basis might work more effectively with a relative lower level of desorptive capacity. Hence, firms should establish their desorptive capacities according to their needs, i.e., external patent exploitation frequency and value creation objective.

7 Conclusion

Along with the open innovation trend, rethinking patent management by considering increased external exploitation of patents to improve the efficiency of the patent portfolio is a crucial aspect for firms. While some pioneering firms use external patent exploitation with great financial and strategic success, many other firms experience substantial managerial difficulties. This paper has explored which factors influence the firms' management and their success of external patent exploitation using a case study approach with fourteen firms from the pharmaceutical and chemical industry. Adopting a desorptive capacity perspective, four main factors were identified: the type of value creation for the firm, the firm's organizational structure, the locus of initiative and the extent of know-how transfer. Furthermore, three archetypes of external patent exploitation with different levels of desorptive capacity were deduced. Specifically, our findings emphasize that creating value through external patent exploitation is not a simple quick win, but requires a certain commitment by the firms in terms of internal effort, collaboration with the partner firm, and

the transfer of know-how. This study is the first which uses the concept of desorptive capacity to analyze the management of external patent exploitation at the firm level. Thus, the article contributes to the concept of desorptive capacity and existing literature on intellectual property management in the context of open innovation. The findings might also help firms to improve their management and success of creating value of their patents by exploiting them outside of the firm boundaries.

Naturally, our exploratory study also has some limitations. First, our study focuses on the pharmaceutical and chemical industry, i.e., industries with discrete technologies. Conducting the study within other industries and different characteristics of the technology, e.g., the semiconductor industry might result in other key factors impacting the firms' approach towards external patent management. Furthermore, we studied mainly European firms. Due to cultural reasons or a development within the open innovation trend that differs from European firms, firms in other countries, e.g., the USA and Japan, might behave differently regarding managing external patent exploitation. Lastly, it has to be taken into account that the generalizability is restricted to due to the number of analyzed firms. Hence, a large scale study testing the influence of the identified factors would be a promising further research path.

The phenomenon of external IP commercialization is becoming increasingly relevant in research and practice, and while existing literature has addressed some issues, many open questions still remain. This paper invites researchers to further engage and investigate white spots in this young research field.

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